

Problem Statement: FinForensic AI

Project Title: FinForensic AI: Institutional Risk & Compliance Terminal

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1. Executive Summary

In the high-stakes environment of institutional finance, manual document auditing is a critical bottleneck. Human analysts struggle to process vast volumes of corporate disclosures, while standard generative AI tools introduce unacceptable risks through "hallucinations." **FinForensic AI** is a multi-agent, deterministic auditing terminal designed to replace manual review with an autonomous, agent-orchestrated governance pipeline.

2. The Problem

- **Operational Inefficiency:** Financial institutions spend thousands of man-hours manually cross-referencing complex regulatory disclosures against institutional mandates.
- **The Hallucination Risk:** Off-the-shelf LLMs lack "grounding" in specific internal documents, leading to unreliable, non-deterministic financial advice.
- **Regulatory Complexity:** Compliance rules change rapidly, and current systems lack a mechanism to proactively block "at-risk" investment strategies before they are executed.

3. The Proposed Solution

FinForensic AI is a reactive, terminal-based platform that utilizes a **"Hard Veto" Agentic Architecture**. Instead of a single chatbot, the system utilizes a "Boardroom" of specialized AI agents:

- **Aggressive Quant Agent:** Analyzes financial returns and market projections.
- **Compliance Officer Agent:** A zero-tolerance watchdog that enforces regulatory guardrails.
- **Macro-Economist Agent:** Synthesizes historical shocks and market trends.
- **Chairman Orchestrator:** Manages the debate between the agents, distilling their conflicting reports into a single, legally defensible, and binding Boardroom Verdict.

4. Innovation & Uniqueness

- **Hard Veto Protocol:** Unlike standard AI, the Compliance Agent has the autonomous authority to trigger a "Hard Veto" if a strategy violates regulatory frameworks (e.g., SEBI guidelines).
- **Deterministic Reasoning:** By leveraging a localized RAG pipeline (ChromaDB), the system forces AI to ground every claim in specific corporate disclosure PDFs, creating an immutable audit trail.

- **Real-Time Telemetry:** The terminal provides "under-the-hood" visibility, allowing auditors to see agent debate lineage and token consumption in real-time via Server-Sent Events (SSE).

5. Technology Stack

- **Backend:** FastAPI (Python) for asynchronous agent orchestration.
- **Frontend:** Angular 18 with RxJS for a reactive, professional terminal interface.
- **AI Engine:** IBM watsonx.ai (Granite-4-h-small model) for high-stakes reasoning.
- **Infrastructure:** Google Cloud Run (Serverless) for deployment; ChromaDB for vector-based knowledge retrieval.